Mass Manufacturing Analysis of RC Car Production

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***Abstract*—This project develops a discrete event simulation of the RC car production system operated in MIT's Laboratory for Manufacturing and Productivity (LMP) to determine how throughput and yearly profit can be maximized under real shop constraints for mass production when scaled to a full year. The goal was twofold: first, to establish a clear baseline of current state performance by accurately modeling all major subassemblies - chassis, drivetrain, suspension, steering, and wheels - using measured cycle time, process plans/machine routing, and a fixed 10 hour daily schedule where each team only receives a 2-hour production window. This baseline allowed us to quantify bottlenecks, utilization, and throughput of each subassembly as well as overall RC Car under existing constraints.**

The second part of the project focused on evaluating improvement scenarios to determine how the system could be optimized for a future production state. We tested changes such as reallocating operators, modifying the layout, and adding capacity through additional equipment. To connect these decisions to financial outcomes, we incorporated the RC Car cost structure where each unit costs $260 to build and generates an $80 profit at a 30% markup. For each scenario, the yearly throughput, resulting profit, and breakeven point were assessed for any capital investment. The analysis showed that waterjet, tapping stations, and hand drill/reamer are the dominant constraints, and that additions in each of these equipment would help improve both throughput and profitability. Overall, the study demonstrates how simulation driven decision making and economic analysis can guide data-driven improvements in constrained multi-part manufacturing systems.

***Index Terms*—RC Car, LMP shop, Utilization, Optimization, Production, Discrete-Event Simulation (DES)**

I. Introduction

Manufacturing systems must balance limited resources, complex process flows, and competing operational constraints to achieve reliable, high-throughput production. Even in small-scale environments, the interaction between shared machines, variable processing times, operator availability, and physical layout can create bottlenecks that significantly impact system performance. This project examines these challenges through the RC car production workflow conducted in MIT's LMP as part of the 2.810 course project. Although the RC car project is instructional in nature, it functions as a realistic multi-part manufacturing system at a small scale. The production of each RC car depended on five different subteams: chassis, drivetrain, steering, suspension, and wheels. Throughout the 2.810, each team designed, manufactured, and assembled various components to manufacture their respective subassembly. Eventually, these subassemblies came together in a final assembly as shown in Fig. 1 below.

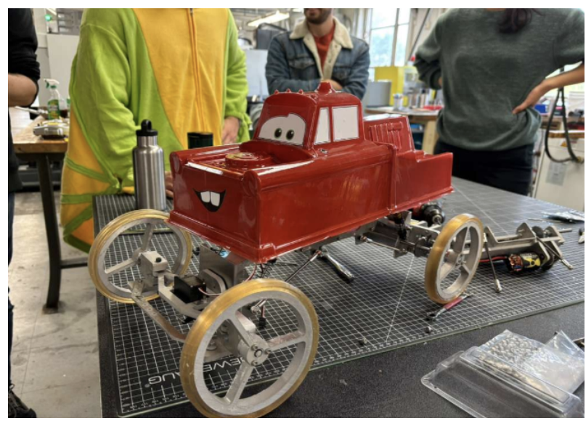

Fig. 1. *Fully Assembled RC Car*

As production ramped up, several qualitative bottlenecks were observed. These inefficiencies prompted the question of how the RC car production can be effectively modeled and optimized within the context of the LMP machine shop. In addition, if this production line were part of an RC car manufacturing business, how much profit could be generated under the current operating conditions, and how could this profit be maximized?

To answer this question, the entire RC car manufacturing process was mapped. Each subteam relies on shared manufacturing resources, such as mills, lathes, drill presses, and the waterjet. The availability of machines in the LMP shop created inherent capacity limitations. The interactions of each subteam, such as operator cycle times and how jobs queue within the 12-hour daily operating window, were studied to develop a holistic model. This model provided insights into what factors are significant in determining the throughput of the system. In addition to operator and job-specific details, the physical constraints of the LMP shop were included in the model as well, such as the location of machines and the dimensions of the LMP shop floor. The combination of physical constraints, machine availability limitations, and operator manufacturing demand all played a critical role in determining feasible production flow and layout options.

The significance of this work lies in its application of core manufacturing systems concepts—factory flow modeling, queuing and utilization, throughput optimization, and economic optimization —to a real, highly constrained production setting. By applying these concepts to a discrete event simulation, changes in personnel allocation and purchases of additional machines were investigated to find the optimal solution to increase the throughput and profitability of the system. These findings were then translated to the context of the 2.810 project to provide actionable recommendations to the LMP staff to facilitate the RC car manufacturing process in future years.

II. Research Question and Motivation

The central question guiding this project is: How can the RC car production system in the LMP shop be modeled, analyzed, and optimized to maximize yearly throughput and profit given strict constraints on machines, labor, and shop operating hours? More specifically, we seek to understand which resources limit production the most, how variability and shared-machine scheduling influence flow, and what combination of layout changes, labor allocations, and equipment investments yields the highest return.

Our motivation for pursuing this project came directly from our experience in 2.810, where producing only 40 RC cars revealed significant bottlenecks, long queues at critical machines, operator overload, and unpredictable delays. Even at this relatively small production volume, it became clear that the system was highly sensitive to machine failures, limited availability of mills and the waterjet, and the fixed 2-hour daily work window per team. These pain points reflected real manufacturing challenges—capacity constraints, unreliable equipment, shared-resource contention, and the difficulty of balancing flow across multiple subassemblies. We wanted to investigate these issues at a deeper, more quantitative level to understand not only what caused the bottlenecks but why they emerged and how they could be mitigated.

This project therefore aligns closely with the core topics from 2.854, including queueing theory, scheduling, variability reduction, reliability modeling, and economic optimization. The RC car factory provides a unique opportunity to apply these concepts to a real, highly constrained system where decisions about staffing, batch sizes, machine redundancy, and layout meaningfully affect performance. By framing the RC car build as a multi-part manufacturing system rather than a class project, we highlight its relevance to real-world factories that must manage limited resources, complex routing, and demand for higher throughput. Through simulation and analysis, our goal is to develop insights that not only improve the RC car workflow but also illustrate robust principles for designing and optimizing small-scale, resource-limited production environments.

III. Adjustments from the Original Proposal

A. Removal of MTTF/MTTR Reliability Model Data

Our original plan included incorporating machine reliability which includes Mean Time to Failure (MTTF) and Mean Time to Repair (MTTR)to better represent downtime and its effect on queueing, operator utilization, and throughput. However, once data collection began, we found that reliable MTTF/MTTR data for the LMP shop machines were unavailable.This challenge is acknowledged in the discrepancy section of our presentation, which notes that reliability data was not captured in the current model.

Given the simulation already featured significant variance from operator differences, cycle-time sampling, and queue interactions, the absence of validated reliability metrics meant that including artificial downtime assumptions risked reducing model accuracy rather than improving it. The team therefore chose to remove reliability modeling from the final simulation, ensuring that results remained grounded in real observed behavior.

B. Exclusion of Explicit Travel-Time Modeling

The original proposal also included modeling operator and part travel time across the machine shop to capture material-handling delays and layout-driven inefficiencies. During layout analysis, we determined that although AnyLogic supports path-based movement modeling, the travel distances in the LMP shop are small relative to cycle times, and most workstations are within a single clustered manufacturing zone.

Because of this, the team assessed travel time as negligible relative to machining, tapping, drilling, and assembly operations. Instead, we emphasized accurate spatial modeling of workstation placement—shown in the original and optimized layouts to ensure that capacity planning and equipment spacing were still represented realistically without the computational overhead of per-operator movement tracking.

Travel time was therefore omitted, but spatial constraints and machine placement were still respected to maintain fidelity in layout-driven decision-making.

IV. Characterization of Available Data

A rigorous analysis of the RC car production system required a comprehensive dataset capable of capturing the true operational behavior of the Laboratory for Manufacturing and Productivity (LMP) shop. Because the goal of the project was to develop a validated discrete-event simulation that reflects real flow dynamics, machine interactions, and resource constraints, the quality and completeness of the input data were critical. To construct this model, we assembled four major categories of data: process cycle times, machine performance and reliability, labor and scheduling parameters, and facility-level layout and spatial information. Each category was sourced from a combination of direct measurement, manual recording during the 2.810 RC car build and asking questions to LMP staff (also 2.810 course instructors).

The first and most essential dataset consisted of measured cycle times for every machining and assembly task across the six major RC car subassemblies. During the 2.810 build, each team recorded their cycle times, including tasks performed on vertical mills, lathes, the waterjet, drill presses, sanding stations, and all downstream assembly steps. Each member was tasked with meeting each of these subteams to get a strong understanding of each other their processes and there bottlenecks might occur. These samples allowed us to estimate empirical distributions rather than relying on simple averages, capturing operator-to-operator variability, tool-change delays, and differences in setup proficiency. Because cycle time variability is a primary driver of queueing and WIP buildup, these measurements formed the foundation of the simulation model.

The second dataset captured labor and scheduling constraints, which are uniquely restrictive in the LMP shop. We set the simulation so that the facility operates for a fixed 12-hour window per day, divided into six 2-hour sessions. This schedule forms a hard upper bound on daily productive time and affects operator availability and machine access. We recorded the number of operators per team, the shared nature of staffing across machines, and the multitasking limitations that prevent continuous supervision of all workstations. These constraints were integrated directly into the simulation to reflect the real bottlenecks caused by limited labor capacity.

Finally, we documented facility layout and spatial constraints, including machine locations, distances between stations, required safety zones, and immovable equipment such as the waterjet (fixed due to plumbing). This information was collected through on-site measurement, inspection of facility blueprints, and consultation with shop staff. These parameters enabled us to incorporate realistic material-handling paths, operator walking distances, and congestion effects—factors that subtly but meaningfully influence throughput.

Throughout the data collection process, we faced several challenges. Cycle-time samples exhibited significant variability due to skill differences, inconsistent setups, and learning-curve effects among student operators. To address this, we retained the full distributions rather than smoothing the data, ensuring the simulation captured real variability. Machine reliability data were difficult to obtain due to the absence of formal logs; we mitigated this by triangulating estimates from staff experience, manufacturer specifications, and observed behavior during the build. Spatial measurements also required manual verification due to discrepancies between actual shop layout and archival diagrams. Despite these challenges, the assembled dataset proved sufficiently representative to construct a validated baseline model and conduct systematic scenario testing.

Taken together, these data sources provided a detailed and realistic characterization of the RC car production system. They allowed us to build a simulation that captures not only the nominal process flows but also the inherent variability, reliability constraints, and human-driven dynamics that define the LMP shop environment. This foundation enabled accurate throughput estimation, bottleneck identification, and economic evaluation of proposed improvements.

TABLE I
CHASSIS PROCESS PLAN AND MEASURED CYCLE TIMES

#	Task	Machine/Tooling	Measured Cycle Time
00	Water jet chassis side plates	Waterjet, calipers	6 mins
01	Tap chassis braces	Tapping arm, M4 tap	2 mins
02	Fixture base plate in milling machine; edge-find first part; use edge stop to locate remaining. Drill 3 holes using conversational milling.	M2.5 drill bit, Milling machine, edge finder	10 mins
03	Flip part 90° and drill two holes; then flip part 180° and drill holes on back side	M2.5 drill bit, Milling machine, edge finder	5 mins
04	Tap all 9 holes on base plate	Tapping arm, M3 tap	3.5 mins
05	Assembly	Allen keys	1.5 mins
		Time / Chassis	**19 mins**

V. SYSTEMS MODELING METHODS CONSIDERED AND CHOSEN

Designing a modeling and analysis approach for the RC car manufacturing system required careful consideration of the system's structure, constraints, and sources of variability. The LMP production environment is characterized by multiple interdependent subassembly flows, shared machining resources, operator-dependent tasks, and significant variability in cycle times and machine reliability. In addition, the shop operates under strict temporal constraints, with only twelve hours of daily operation divided into six two-hour production windows. These conditions create a complex, stochastic, and highly interactive manufacturing system—one that cannot be accurately represented through simple deterministic or spreadsheet-based calculations. As such, our team evaluated several potential modeling methods before selecting a discrete-event simulation (DES) as the most appropriate and powerful tool for this project.

Initially, we explored the possibility of building a capacity model in Excel, leveraging average cycle times and basic queueing approximations to estimate throughput. While Excel provided a quick and accessible way to visualize the process flow and calculate rough utilization levels, it lacked the ability to capture the dynamic behavior of queues, operator-sharing logic, random machine failures, or process variability. Because the RC car factory's performance is driven heavily by the interaction between stochastic events—such as variable cycle times and downtime—and resource contention, a purely deterministic spreadsheet model would have risked oversimplifying the system and producing misleading results.

We next considered building a custom simulation using Python (via SimPy) or MATLAB (via SimEvents). These tools offered more flexibility and allowed for discrete-event modeling, but they introduced significant development overhead.

Constructing the fundamental logic from scratch—resource objects, queues, failure events, operator assignment rules, routing dependencies, and layout constraints—would have required substantial time and debugging effort. Moreover, these platforms lacked intuitive, built-in visualization, making it difficult to validate the model or communicate results to stakeholders. Given our project timeline and the need to run numerous what-if scenarios, the trade-offs of a fully custom-coded simulation did not justify the potential benefits.

Machine learning techniques were also briefly considered, particularly for predicting cycle-time variation or failure likelihood. However, ML approaches rely on large historical datasets, which do not exist for the RC car build, and they cannot natively model the system-level dynamics necessary for evaluating throughput, WIP behavior, or resource interactions. While ML remains an attractive extension for future work, it was not suitable as a primary modeling method for this project.

Ultimately, we selected discrete-event simulation using AnyLogic as our primary modeling and analysis tool. DES is uniquely suited to representing manufacturing systems in which events occur at discrete points in time—such as job arrivals, machine start/finish times, failures, repairs, and operator handoffs. AnyLogic provided a comprehensive simulation environment capable of integrating all key aspects of the LMP shop: stochastic cycle-time distributions, shared machines and operators, downtime modeling via MTTF/MTTR, routing across subassembly lines, and even approximate spatial layout and travel distances. The platform also allowed us to animate the system in real time, making it far easier to detect modeling errors, validate assumptions with LMP staff, and demonstrate bottlenecks visually.

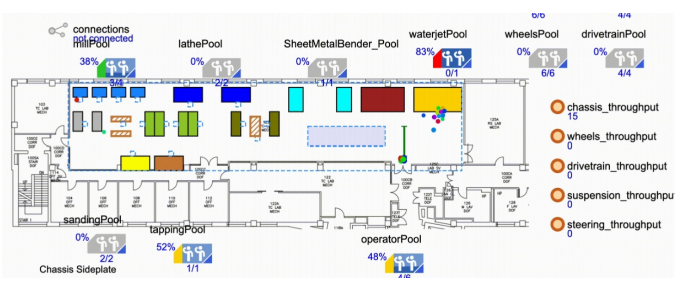

Fig. 2. *LMP simulation to simulate walkways and machine placement in Anylogic*

The use of AnyLogic also enabled rapid experimentation with different scenarios. We were able to test alternative staffing rules, evaluate the impact of machine additions such as a second waterjet or extra mills, modify buffer locations and batch sizes, and introduce layout variations—all without rebuilding the model structure. This flexibility was essential for performing economic analysis, as it allowed us to compare the throughput and profit implications of each capital investment option.

In summary, discrete-event simulation was selected because it is the only modeling method capable of fully capturing the dynamic, stochastic, and interactive behavior of the RC car manufacturing system. Its ability to represent queues, variability, shared resources, downtime events, and layout constraints made it the most effective tool for analyzing bottlenecks and evaluating improvement strategies. The use of AnyLogic allowed us to build a validated digital representation of the LMP shop and perform a rigorous set of scenario analyses, ultimately guiding our recommendations for maximizing throughput and profit under the existing operational constraints.

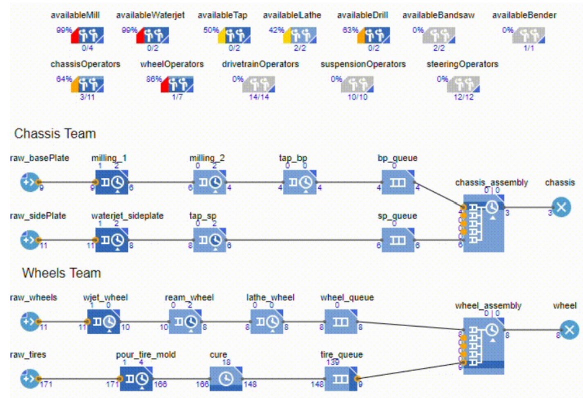

Fig. 3. *serial processes needed for the chassis and wheels team to create one assembly. Each processes will only be processed when an available equipment and an available operator is ready*

VI. Empirical Findings from Modeling

This section presents the results of the discrete-event simulation using Anylogic and examines how the current production system behaves over a one-year operating period. The analysis focuses on **subassembly throughput, resource utilization**, and the **identification of the primary bottleneck** limiting RC-car output.

A. Throughput of Subassemblies

After simulating one full year of work under the current setup, we found the following yearly outputs for each subassembly team:

TABLE II
Subassemblies Required to Build One Car per Year

Subassembly Teams	Subassemblies per Year
Chassis	2224
Wheels	1953
Drivetrain	2157
Suspension	3050
Steering	1640

Each RC car requires a different number of parts from each team (for example, we need four wheels, two suspension assemblies, and one chassis). When we adjust for these quantities, we found that steering is the slowest team and limits how many full cars can be produced.

Because of this, improving **steering** output became the primary focus of our optimization efforts. If steering can

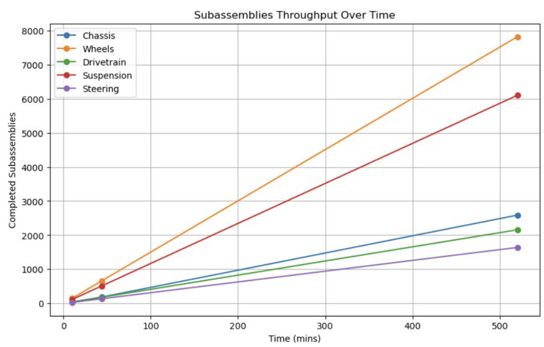

Fig. 4. *Sub-assemblies Throughput Over Time*

increase its throughput, the entire system's output can increase as well.

B. Utilization in Subassemblies

These measures indicate how frequently each resource is actively engaged in production and help reveal whether bottlenecks arise from insufficient labor, inadequate machine capacity, or both. The utilization values for other subassemblies are provided in the Appendix A.

TABLE III
EQUIPMENT QUANTITY AND UTILIZATION RATES

Equipment	Quantity	Utilization Rate (%)
Milling Station	4	37
Tapping Station	1	40
Waterjet Station	1	99
Sanding/Bandsaw Station	1	7
Members	6	62

The utilization results show that the waterjet station is the main bottleneck for the steering team. With a 99% utilization rate, the waterjet is essentially running at full capacity all year, which restricts how many steering parts can be produced. Meanwhile, operator utilization is only 62%, meaning members spend a significant amount of time waiting for parts to finish on the waterjet. Other machines— including the mill and sanding/bandsaw—have much lower utilization because work cannot move to these steps until the initial waterjet cuts are complete. This fits the structure of the steering workflow, as all major components (Rigid Plate Mount, Rigid Linkage, and Steer Bars) start on the waterjet. As a result, the waterjet ultimately determines the output of the entire steering line.

C. Bottleneck Score Analysis

In order to understand where the RC car production system is actually becoming constrained, we combined each machine's utilization with the throughput of its subassembly. Raw utilization alone only indicates how busy a machine is, but not how that usage limits completed cars. To address this, we calculated a bottleneck score, which normalizes utilization by throughput:

$$\text{Bottleneck Score} = \frac{\text{Machine Utilization (\%)}}{\text{Subteam Throughput}} \tag{1}$$

This approach helped us identify which machines meaningfully restrict output versus those that simply operate at high utilization due to lighter workloads or faster cycles. For example, while several machines run close to full capacity, the bottleneck score showed that the waterjet is the dominant constraint across the system, affecting four out of the five teams. Steering, in particular, has the highest bottleneck score (0.060366), confirming that it is the largest limiter of complete RC car throughput and the top priority for any improvements.

Something we found particularly interesting was that Drivetrain's limiting resource was not the waterjet but the tapping station, which only becomes apparent after adjusting for throughput. Insights like this guided where additional machines (e.g., waterjet, tapping) would create the largest system-wide gains. Overall, bottleneck scores provided a more complete picture than utilization alone and allowed us to make targeted, high-impact decisions on capacity investments.

TABLE IV
PRIMARY BOTTLENECKS AND BOTTLENECK SCORES BY SUBTEAM

Subteam	Primary Bottleneck Machine	Bottleneck Score
Chassis	Waterjet	0.038268
Wheels	Waterjet	0.012670
Drivetrain	Tapping	0.028280
Suspension	Waterjet	0.016230
Steering	Waterjet	0.060366

D. Optimization through Machine Purchases

After establishing how the current state of the LMP shop performs through examining the subassembly throughput, machine utilization, and bottleneck scores, the natural next step was to evaluate how these constraints could be optimized. Our guiding question required us to understand not only how the current factory layout operates, but also what targeted changes would most effectively increase overall RC car throughput. The optimization work therefore centered on using data from the simulations to guide decisions on purchasing additional machines or adding operators.

The bottleneck score analysis made it immediately clear that Steering is the primary constraint on RC car output. Even though four out of five subteams are constrained by the Waterjet, Steering's bottleneck score is the highest, and its subassembly throughput is the lowest at roughly 1,640 completed units. Since the RC car throughput can never exceed the throughput of its slowest subassembly, steering became our starting point for optimization.

We first compared the cost and impact of adding machines. Steering's utilization data showed a 99% Waterjet utilization, which indicated that the Waterjet was the controlling constraint. We therefore added one Waterjet in the model and reran the simulation. This single change resulted in a substantial increase in Steering throughput, and as expected, Waterjet utilization decreased. The improvement also financially justified the investment: with each RC car earning a $70 net profit and a Waterjet costing approximately $27,500, the increase in system throughput more than offset the machine cost.

Once the Waterjet constraint was relieved, the next limiting factor shifted to team staffing. We increased the Steering team size from six to seven members, reran the simulation, and observed another rise in throughput. This iterative "stepping" approach allowed us to identify the point at which additional machines or people no longer provided proportional improvements. The table below shows an example of the stepping approach of steering, with changes highlighted in green and bottlenecks highlighted in red.

Across these runs, the most impactful and practical optimum for Steering was achieved by adding one Waterjet and one additional team member, which increased Steering throughput to 2708 units. This is because we need to consider space constraints at the LMP when adding an additional waterjet and operator availability for other subteams as well. While not the absolute maximum explored, this configuration balanced cost, machine loading, and operator availability without pushing the system into diminishing returns. The corresponding changes in throughput, net profit, and utilization are shown in the figures below:

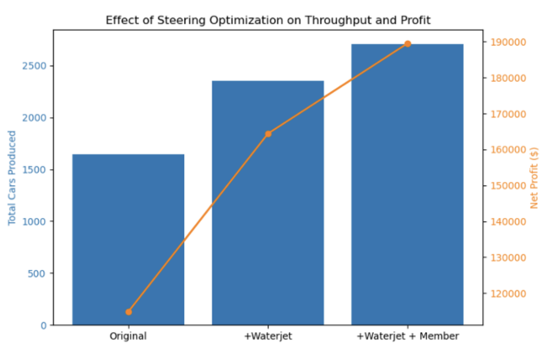

Fig. 5. *Steering Optimization Effects on Throughput, Profit, and Machine Utilization*

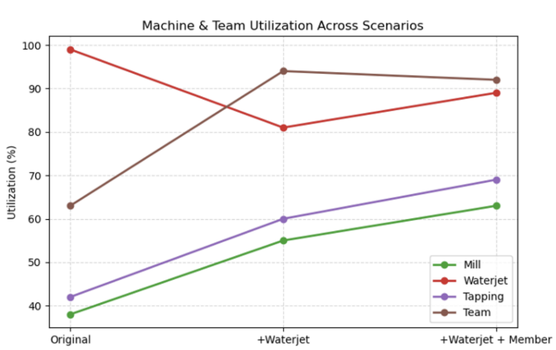

Fig. 6. *Machine and Team Utilization*

This same stepping approach was repeated for Wheels, Drivetrain, Chassis, and Suspension. Although each team had its own local constraints, the pattern was consistent: Waterjet and tapping capacity most frequently limited throughput, and adding small amounts of additional capacity produced meaningful improvements. Across the full factory, the set of machine investments that generated the most balanced performance is shown in the table below:

TABLE V
AVAILABLE MACHINES AND COUNTS IN THE LMP SHOP

Machine	Count
Mill	4
Waterjet	2
Tap	2
Lathe	2
Reamer / Hand Drill	2
Bandsaw	2
Sheet Metal Bender	1

By adding 1 Tap, 1 Waterjet, and 1 Hand Drill/Reamer, all subassemblies benefited from reduced machine utilization. The "stepping" response of the other subassemblies are shown in the Google Sheet in Appendix A under the "Stepping Data" tab attached to the appendix.

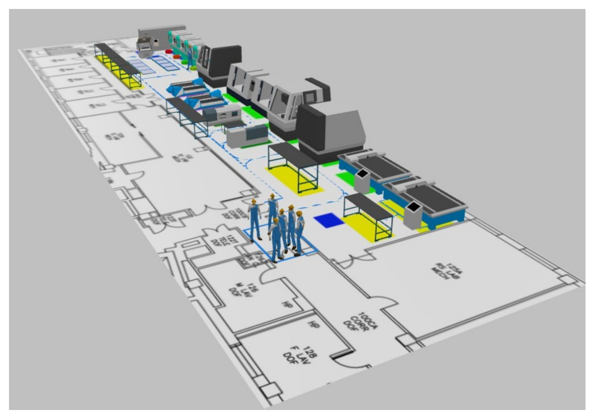

Fig. 7. *Optimized LMP Layout show in AnyLogic*

E. Optimization through Operator Allocation

Once the machine mix was optimized, the next question was how to allocate operators across the five subassemblies to create a balanced and efficient workflow. The goal here was straightforward: with a fixed maximum number of operators available, determine the allocation that maximizes overall RC car throughput while keeping each team appropriately utilized. We started by looking at the current operator utilization levels from the simulation. The figure below shows the operator utilization by subteam for the unoptimized machine model:

This showed two immediate issues:

- **Drivetrain and Chassis** are heavily utilized, meaning they operate close to capacity and risk becoming future bottlenecks.
- **Wheels and Steering** appear underutilized, but Steering's utilization is misleadingly low because it was previously limited by machine capacity rather than people.

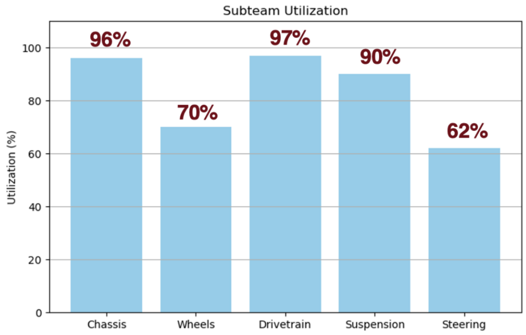

Fig. 8. *Subteam Utilization*

With machine bottlenecks removed in the previous stage, we now held the machine configuration constant and varied only the number of operators. Using AnyLogic, we swept through operator counts from 1 to 20 for each subassembly and measured how throughput changed. This made it easy to see the point at which additional operators stopped increasing output, essentially showing the saturation points and/or diminishing returns. The figure below shows the throughput varying by operator count for each subassembly:

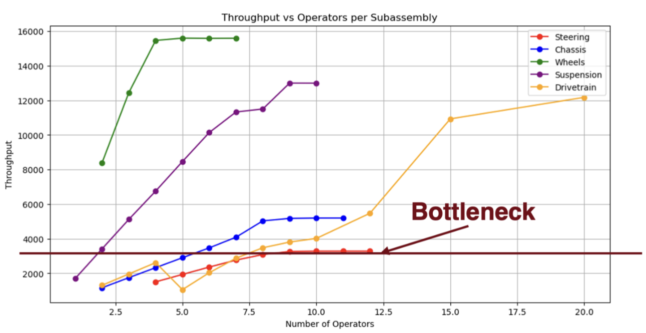

Fig. 9. *Subteam Utilization*

The results confirmed again that Steering remains the system bottleneck, even with improved machine capacity. Every additional operator given to Steering had a clear and measurable impact on total RC car throughput, whereas increasing operators on teams like Wheels or Suspension quickly yielded no improvement. With a constraint of 24 total operators, our task became allocating people in a way that ensures all subassemblies produce at roughly the same level as Steering's maximum sustainable throughput. We found the best performing operator allocation to be:

This allocation strikes the right balance across all subassemblies while staying within the 24 operators limit. Most importantly, it allows Steering to reach a peak throughput of 2,765 steering subassemblies per year, which becomes the new optimized limit for complete RC car production. All other teams, with the revised operator allocation, are able to produce at, or slightly above, this level, maintaining a balance in subassembly production.

TABLE VI
OPTIMIZED OPERATOR ALLOCATION BY SUBTEAM

Subteam	Operators	Change
Drivetrain	7	+3
Steering	7	+1
Chassis	5	+1
Wheels	3	-3
Suspension	2	-2

F. Comparison of Optimal Parameters versus Current Parameters

One of the initial questions driving this study was to compare the total profit generated using both the original and optimized production structures. An analysis of the bill of materials of each subteam was conducted to understand the total raw materials cost required to build each subassembly. A summary of each subassembly's cost is shown in table x below.

TABLE VII
MANUFACTURING COST BY SUBASSEMBLY

Subassembly	Cost ($)
Steering	25.48
Suspension	55.96
Chassis	16.75
Drivetrain	66.11
Wheels	39.23
TOTAL Manufacturing Cost	**203.53**

The total manufacturing cost to complete 1 RC car is $203.53. Industry research suggests that an average RC car is sold for a 30% profit margin. Therefore, establishing a 30% profit margin means that each car sold would contribute $61.06 in profits.

In an unchanged production plan, 1640 RC cars can be manufactured in a year. Multiplying this value by the profit per RC car yields a total profit of $100,138.40. The optimized line produced 2765 RC cars, generating $168,830.90 in gross profits. However, the optimized line also involved changes in personnel allocations between teams, as well as the purchase of additional equipment. Changing the number of operator per subteam did not contribute any extra cost since the total number of people involved in the production line remains the same. The cost penalty for purchasing additional machines is shown in table x below. These prices were found through contacting the vendors of similar equipment.

In the case of the optimized line, an additional waterjet cutter, hand drill, and tapping machine were purchased, resulting in a total purchasing cost of $28,800. Subtracting this sum from the gross profit results in a net profit of $140,030.90. The optimized line produced $39,892.50 in additional profits compared to the original production line. Therefore, the reallocation of personnel and the purchasing of additional equipment were justified financial decisions to generate additional profit.

TABLE VIII
COST OF MACHINES USED IN PRODUCTION

Machine	Price ($)
Manual Mill (Bridgeport)	17,000
Waterjet Cutter (OMAX 2652)	27,500
CNC Lathe (HAAS TL-1)	35,995
Hand Drill	200
Bandsaw (DoALL Vertical Bandsaw)	17,000
Sander (JET Vertical Combo)	1,649
Tapping	1,100

VII. COMPARISON TO REAL DATA OR ORIGINAL METHOD

Overall, the simulation model served as a strong approximation of how the process and parts actually flows through the shop. Many of the trends we observed in the model aligned closely with our own intuition and prior experience in building these cars at the LMP shop. For example, the model immediately showed the Waterjet as a dominant bottleneck, which was consistent with what we saw during manufacturing. Since the LMP only has its own waterjet and almost every subassembly starts with cutting stock metal through the waterjet, every team waits on it, building a queue, and any delays stall multiple subassemblies. Similarly, the model correctly identified the tapping bottleneck for the Drivetrain subbasembly. 3 out of the 4 parts of the drivetrain assembly needed to be tapped which caused backlog and queues to form very quickly given the shop only has one tapping station. Moreover, even the general ranking of subassembly throughput/difficulty was accurate: Steering and Drivetrain consistently came out slowest, Wheels and Suspension fastest, and Chassis placed somewhere in the middle.

However, there are a few important areas where the model simplifies or misses the real-world behavior. For example, the biggest limitation is operator variability. The model assumes perfectly consistent task times without any interruptions or workflow transitions. In reality, teams will differ in experience, causing few members to specialize in a particular task, limiting availability. Additionally, the model does not include setup activities such as homing the waterjet, tool changes, fixture alignment, which accumulate across the day, significantly affecting throughput. The physical shop layout is also not fully represented. Given the small size of the LMP shop, this was accounted to be negligible in comparison to the cycle times of each workstation. However, incorporating the walking time, machine proximity, and any type of workstation congestion could more accurately quantify the throughput. Finally, machine reliability was not modeled due to incomplete MTTF/MTTR data in the shop logs. As it was noticed during manufacturing of the actual RC car, downtime events, especially for the Waterjet given its reliance and singular nature, can significantly impact actual output and were not incorporated into this version of the simulation.

Overall, the model captures the core dynamics well such as the bottlenecks, queues, and resource availability, but remains an idealized version of the real factory, with clear opportunities for improvements.

VIII. RECOMMENDATIONS FOR FUTURE WORK

A. Data Expansion for Accuracy

Collect more cycle times with different operators

In our data collection, we recorded only ten cycle-time samples using the operators who were available at the time. We then generalized these measurements by assuming that all operators share the same mean and variance. In reality, operators differ in skill level, training, and familiarity with each machine. Collecting cycle times from a broader range of operators would allow us to capture these differences and identify which team members perform best on specific machines. This information could help guide more effective operator assignments in future studies.

During real observations, we noticed that teams prioritize different parts of the build depending on the day. For example, on the first day, one operator may fully occupy the waterjet to produce all required cut parts. On the following day, the team may focus on secondary processes such as milling or tapping, leaving the waterjet with little to no use. Incorporating these shifting priority behaviors into the model would help replicate actual workflow patterns more accurately and may reduce artificial bottlenecks by ensuring that machine usage aligns more closely with team needs and project urgency.

B. Model Enhancement

Account for setup times during team transitions. Each team operates within a two-hour work window, which introduces setup time whenever processes are started or switched between teams. These setup activities—including fixturing, tool changes, and machine preparation—reduce effective production time and vary by machine. Incorporating setup times would improve the accuracy of cycle-time and utilization estimates.

Include lead times for material procurement.

Some components are sourced from suppliers outside of McMaster-Carr, resulting in delays that are not currently captured in the model. Adding material lead times would better reflect real scheduling constraints and improve throughput predictions.

Incorporate machine reliability metrics.

Observed machine failures during the course, such as broken taps and extended waterjet downtime, highlight the need to model equipment reliability. Including Mean Time to Failure (MTTF) and Mean Time to Repair (MTTR) would allow the simulation to account for downtime and better represent real operating conditions.

C. Improved Optimization

Optimize operation sequencing.

Certain parts allow flexibility in the order of operations. For example, performing drilling and tapping before facing may reduce competition for shared machines and help distribute workload more evenly across the shop. Evaluating alternative operation sequences could reduce queueing at bottleneck processes and improve overall flow.

Evaluate alternative designs, materials, and manufacturing methods.

Design choices directly affect manufacturing effort and machine usage. Some components could be redesigned to use less-constrained processes, such as cutting chassis holes on the waterjet instead of milling, which would reduce operator movement and setup time. Modeling these alternatives would clarify trade-offs between throughput, cost, and process selection and support more informed design decisions.

IX. Conclusion

Overall, our analysis shows a clear improvement between the current factory setup and the optimized configuration we developed. In the baseline scenario, the system could only produce around 1,640 RC cars per year, yielding an estimated net profit of $100,138.40. After optimizing both machine allocations and operator assignments, the model achieved a maximum throughput of 2,765 parts per year and an optimized profit of $140,030.90, which is an overall increase of $39,892.50. This represents a substantial improvement in both productivity and economic performance, driven by identifying and relieving key bottlenecks, especially within Steering.

At the same time, it is important to recognize that this remains a model, and the real-world system includes factors that are harder to perfectly capture such as machine reliability, setup times, variability in operator skill, and daily disruptions. Translating our model-based learnings into practical recommendations for the LMP shop, we highlight that the Waterjet remains the dominant constraint across multiple subassemblies. Given the physical layout and access limitations, a realistic near-term solution would be for LMP to partner with another lab that has a waterjet rather than purchasing and installing another unit internally. Additionally, student allocation across subteams should be revisited for the 2.810 course to ensure that operator capacity is aligned with true bottlenecks and overall flow.

In summary, the optimized configuration demonstrates how targeted machine and operator adjustments can significantly improve throughput, reduce congestion, and enhance overall productivity. While the model simplifies the real factory environment, the insights provide a strong foundation for future improvements in both LMP operations and the 2.810 learning experience.

Appendix A
Data Source

https://docs.google.com/spreadsheets/d/1eD0l6eBGSqJiVMTS7Mhe6sn269kXRhC9bTCfmEPHn64/edit?usp=sharing